

Phase 7: Integration & External Access

Goal:

Enable Salesforce to securely communicate with external systems (fraud detection API, other services) while ensuring data security, scalability, and compliance with Salesforce governor limits.

1. Named Credentials (✓ Used)

Purpose:

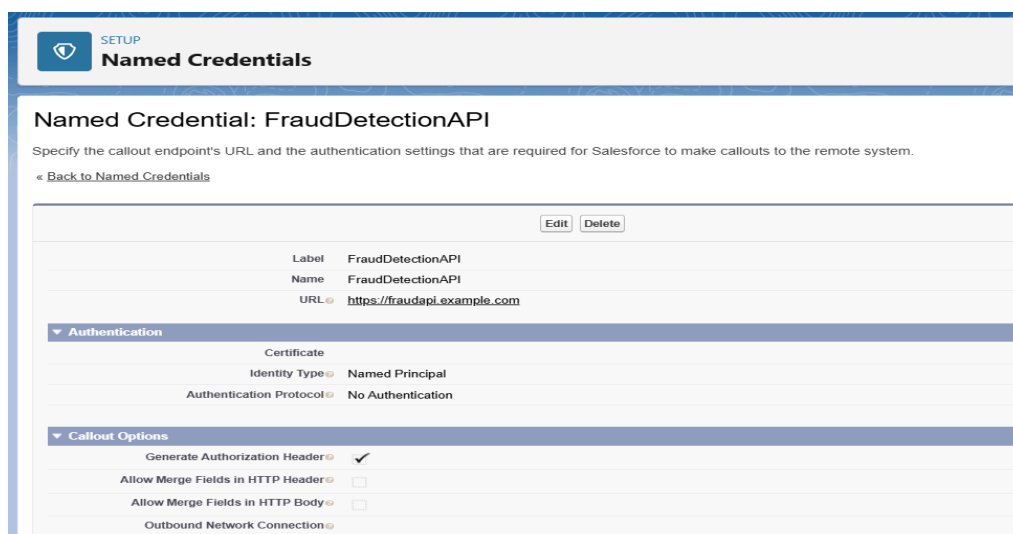
Securely store the base URL and authentication settings for an external API without hardcoding credentials inside Apex classes.

Steps Followed:

1. Setup → Named Credentials → New.
2. Added **Label: FraudAPI** and **URL: https://fraudapi.example.com**.
3. Chose **Identity Type: Anonymous** and **Authentication Protocol: No Authentication**, since using a demo API without login credentials.
4. Saved the configuration.

Why Important?

- Avoids hardcoding API endpoints in Apex.
- Centralized management of authentication.
- Secure and easy to update if endpoint changes.



The screenshot shows the Salesforce Setup interface for configuring a Named Credential. The page title is "Named Credentials" with a "SETUP" link. Below the title, it says "Named Credential: FraudDetectionAPI". A subtitle reads: "Specify the callout endpoint's URL and the authentication settings that are required for Salesforce to make callouts to the remote system." There is a link to "Back to Named Credentials".

The configuration form includes the following fields:

- Label:** FraudDetectionAPI
- Name:** FraudDetectionAPI
- URL:** https://fraudapi.example.com

Below these fields are two expandable sections:

- Authentication:** This section is expanded, showing:
 - Identity Type:** Named Principal
 - Authentication Protocol:** No Authentication
- Callout Options:** This section is also expanded, showing:
 - Generate Authorization Header:** ☒
 - Allow Merge Fields in HTTP Header:** ☐
 - Allow Merge Fields in HTTP Body:** ☐
 - Outbound Network Connection:** ☐

2. External Services (❌ Not Used)

Purpose:

Allows Salesforce to consume APIs described by an OpenAPI (Swagger) specification without writing code. Salesforce auto-generates Apex actions from the service schema.

Reason for Skipping:

- Our demo fraud detection API doesn't provide Swagger/OpenAPI definitions.
- Instead, we directly integrated via Named Credentials + Apex Callout.

3. Web Services (REST/SOAP) (✔ Used – REST)

Purpose:

Salesforce supports integrating with external systems via REST or SOAP APIs.

- REST → lightweight, JSON-based (used in our project).
- SOAP → XML-based, older format (not used here).

Implementation:

- Created an Apex class with `@future(callout=true)` to send REST callouts.
- Example: Fetch fraud score for a claim using `req.setEndpoint('callout:FraudAPI/score')`.

Code:

```
public with sharing class FraudAPIService {

    // Future method for async callout
    @future(callout=true)
    public static void updateFraudScore(Id claimId) {
        try {
            // Get claim record
            Insurance_Claim__c claim = [
                SELECT Id, Claim_Number__c
                FROM Insurance_Claim__c
                WHERE Id = :claimId
                LIMIT 1
            ];
```

```

Http http = new Http();
HttpRequest req = new HttpRequest();

// Use Named Credential → FraudAPI_NC (created earlier)
req.setEndpoint('callout:FraudAPI_NC/checkFraud?claimId=' + claim.Claim_Number_
_c);
req.setMethod('GET');

HttpResponse res = http.send(req);

if (res.getStatusCode() == 200) {
    // Parse JSON response
    Map<String, Object> result =
        (Map<String, Object>) JSON.deserializeUntyped(res.getBody());

    if (result.containsKey('fraudScore')) {
        Decimal score = Decimal.valueOf(result.get('fraudScore').toString());

        // Update claim with fraud score
        claim.Risk_Score__c = score;
        update claim;
    }
} else {
    System.debug('Fraud API Callout failed → ' + res.getStatusCode() + ' : ' + res.getBo
y());
}
} catch (Exception e) {
    System.debug('Error in FraudAPIService.updateFraudScore: ' + e.getMessage());
}
}
}

```

Apex Classes

Apex Class

FraudAPIService

Help for

Apex Class Detail

Edit

Delete

Download

Security

Show Dependencies

Name	FraudAPIService	Status	Active
Namespace Prefix		Code Coverage	0% (0/17)
Created By	Sanjivani Dashmukh , 9/23/2025, 6:20 AM	Last Modified By	Sanjivani Dashmukh , 9/23/2025, 6:38 AM

Class Body

Class Summary

Version Settings

Trace Flags

```

1 public with sharing class FraudAPIService {
2
3     // Future method for async callout
4     @future(callout=true)
5     public static void updateFraudScore(Id claimId) {
6         try {
7             // Get claim record
8             Insurance_Claim__c claim = [
9                 SELECT Id, Claim_Number__c
10                FROM Insurance_Claim__c
11                WHERE Id = :claimId
12                LIMIT 1
13            ];
14
15            Http http = new Http();
16            HttpRequest req = new HttpRequest();
17
18            // Use Named Credential → FraudAPI_NC (created earlier)
19            req.setEndpoint('callout:FraudAPI_NC/checkFraud?claimId=' + claim.Claim_Number__c);
20            req.setMethod('GET');
21
22            HttpResponse res = http.send(req);

```

4. Callouts (✔ Used)

Purpose:

Allow Salesforce to send HTTP requests (GET/POST) to external systems.

Steps:

1. Added Named Credential → FraudAPI.
2. Wrote Apex code:
3. `HttpRequest req = new HttpRequest();`
4. `req.setEndpoint('callout:FraudAPI/score');`
5. `req.setMethod('GET');`
6. `HttpResponse res = new Http().send(req);`
7. Parsed JSON response and updated Claim Risk Score.

Why Important?

- Enables Salesforce to interact with real-time fraud detection services.
- Provides fraud score enrichment for Insurance Claims.

SETUP

Remote Site Settings

Remote Site Details

Remote Site Detail

Edit

Delete

Clone

Remote Site Name

FraudAPI

Modified By

Sanjivani Deshmukh

9/23/2025, 6:33 AM

Remote Site URL

https://fraudapi.example.com

Disable Protocol Security

☐

Description

Active

☒

Created By

Sanjivani Deshmukh

9/23/2025, 6:33 AM

Edit

Delete

Clone

5. Platform Events (Used)

Purpose:

Enable event-driven architecture for real-time communication inside Salesforce and with external systems.

Example in Our Project:

- Created a Fraud_Alert__e platform event.
- Published event when claim fraud score > 80.
- External listeners can react to fraud alerts immediately.

Why Important?

- Decouples fraud detection logic from notifications.
- Supports real-time alerting and system integration.

SETUP

Platform Events

Platform Event

FraudAlertEvent

Standard Fields (4)

Custom Fields & Relationships (4)

Platform Event Definition Detail

Edit

Delete

Singular Label

FraudAlertEvent

Description

Plural Label

FraudAlertEvents

Deployment Status

Deployed

Object Name

FraudAlertEvent

API Name

FraudAlertEvent__e

Event Type

High Volume

i

Publish Behavior

Publish Immediately

i

Created By

Sanjivani Deshmukh

9/23/2025, 6:46 AM

Modified By

Sanjivani Deshmukh

9/23/2025, 6:46 AM

Standard Fields

Action	Field Label	Field Name	Data Type	Controlling Field	Indexed
	Created By	CreatedBy	Lookup(User)		
	Created Date	CreatedDate	Date/Time		
	Event UUID	EventId	Text(36)		
	Replay ID	ReplayId	External Lookup		

Custom Fields & Relationships

New

Action	Field Label	API Name	Data Type	Indexed	Controlling Field	Modified By	
Fri Del	ClaimNumber	c	ClaimNumber	c	Text(20)	Sanjivani Deshmukh	9/23/2025, 6:47 AM

```
File Edit Debug Test Workspace Help < >
FraudAlertPublisher.apxc
Code Coverage: None API Version: 64
1 public class FraudAlertPublisher {
2
3     // Method to publish the Fraud Alert Event
4     public static void publishFraudAlert(String claimNumber, Decimal fraudScore, String customerName) {
5
6         // Create an instance of your Platform Event
7         FraudAlertEvent__e alert = new FraudAlertEvent__e(
8             ClaimNumber_c__c = claimNumber,
9             FraudScore_c__c = fraudScore,
10            CustomerName_c__c = customerName,
11            Message_c__c = 'High-risk claim detected'
12        );
13
14        // Publish the event
15        Database.SaveResult result = EventBus.publish(alert);
16
17        if (result.isSuccess()) {
18            System.debug('Fraud Alert Event Published Successfully: ' + claimNumber);
19        } else {
20            System.debug('Failed to publish event: ' + result.getErrors()[0].getMessage());
21        }
22    }
23 }
```

 **Apex Triggers**

Apex Trigger

FraudAlertEventTrigger

Apex Trigger Detail

Edit Delete Download Show Dependencies

Name	FraudAlertEventTrigger	sObject Type	FraudAlertEvent
Code Coverage	0% (0/1)	Status	Active
Created By	Sanjivani Deshmukh, 9/23/2025, 7:07 AM	Last Modified By	Sanjivani Deshmukh, 9/23/2025, 7:13 AM
Namespace Prefix			

Apex Trigger

Version Settings Trace Flags

```
1 trigger FraudAlertEventTrigger on FraudAlertEvent__e (after insert) {
2     for(FraudAlertEvent__e eventRecord : Trigger.New) {
3         System.debug('Fraud Alert for Claim: ' + eventRecord.ClaimNumber_c__c);
4         // Call external system via Apex callout if needed
5     }
6 }
```

Edit Delete Download Show Dependencies

6. Change Data Capture (❌ Not Used)

Purpose:

Automatically publishes events whenever records in selected objects change.

Reason for Skipping:

- Not essential for our demo project.
 - Could be used in future for syncing Claim/Investigation updates with external databases.
-

7. Salesforce Connect (❌ Not Used)

Purpose:

Connects external databases (like Oracle, SAP, AWS) as virtual objects in Salesforce without storing data locally.

Reason for Skipping:

- Our fraud detection service provides API access only, not database connection.
 - Salesforce Connect not required.
-

8. API Limits

Purpose:

Salesforce imposes limits to ensure system performance.

Important Limits:

- **REST API Calls per 24 hours** → Based on license (e.g., 15,000 for Enterprise Edition).
- **Callout timeout** → 120 seconds max.
- **Concurrent callouts** → Up to 10 per Apex transaction.

Why Important?

- Helps us design integrations without exceeding limits.
 - In real projects, ensures API quota is not exhausted.
-

9. OAuth & Authentication (❌ Not Used – Chose No Auth)

Purpose:

OAuth2 is a standard authentication method for connecting Salesforce to third-party services.

Why Skipped:

- Our fraud detection API is a demo and doesn't require credentials.
- Chose **No Authentication** in Named Credentials.

Note:

In a real system, APIs usually require OAuth2 / Password Authentication for security.

10. Remote Site Settings (✔ Used)

Purpose:

Allow Salesforce to call external services securely.

Steps:

- For Named Credentials → Remote Site Settings are automatically handled.
- If calling endpoint directly, you must manually add base URL in **Setup → Remote Site Settings**.

Why Important?

- Prevents unauthorized external callouts.
- Adds a layer of security by whitelisting allowed domains.

The screenshot shows the 'Remote Site Settings' page in Salesforce. At the top, there is a 'SETUP' icon and the title 'Remote Site Settings'. Below this, the 'Remote Site Details' section is visible. It contains a table with the following information:

Remote Site Detail		Edit	Delete	Clone
Remote Site Name	FraudAPI			
Remote Site URL	https://fraudapi.example.com			
Disable Protocol Security	☐			
Description				
Active	☒			
Created By	Sanjivani Deshmukh, 9/23/2025, 6:33 AM			

At the bottom of the table, there are buttons for 'Edit', 'Delete', and 'Clone'.